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ARTICLE

Efficiency of forest carbon policies at intensive and extensive margins

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Abstract
The economic potential of forest carbon sequestration is widely acknowledged. However, no consensus has been reached regarding the appropriate policy instrument for promoting carbon sequestration. In this study, we develop a dynamic framework to measure the effects and efficiencies of alternative carbon policies. A stylized optimal control model of the timber market is first employed to illustrate the mechanisms through which different policies affect the decision making of the forest sector at the extensive margin (i.e., changing forest areas) and the intensive margin (i.e., changing harvest ages). We then introduce carbon price projections and species-specific production information into a multi-age dynamic timber market model. Different carbon policies are simulated numerically. Our results reveal that a carbon tax on forest emissions without compensating for sequestration leads to net carbon emissions and, thus, is the least efficient policy choice. Further, policies that do not increases carbon uptake at the intensive margin result in very high efficiency losses. A per-hectare land subsidy may be more than 10 times more expensive than a per-ton carbon tax and subsidy policy or a carbon subsidy policy.

KEYWORDS
carbon policy, carbon sequestration, dynamic model, forest

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1 | INTRODUCTION

Forest carbon sequestration can play a critical role in the global carbon budget. Following the Paris Agreement on emissions targets to limit the global temperature increase in this century to 2 degrees

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Celsius or even further to 1.5 degrees, many countries have incorporated emissions and sequestration projections from land use, land-use change, and forestry (LULUCF) activities into their intended nationally determined contributions (INDCs), which become nationally determined contributions NDCs as countries formally join the Paris Agreement. There are 68 countries (or 41 NDCs, with the EU's NDCs representing its constituent 28 countries) with specific LULUCF contributions in their total mitigation targets, including both unconditional and conditional targets, corresponding to around 78% of global net emissions in 2012 and 83% of the global forest area (Grassi et al., 2017). It is believed that additional countries are also considering the potential role of forests and forest products in their mitigation targets beyond the next 10 to 15 years (Forsell et al., 2016; Rogelj et al., 2016).

Although the economic potential of forest carbon sequestration is widely acknowledged (Golub et al., 2009; Plantinga et al., 1999; Richards & Stokes, 2004; Sohngen & Mendelsohn, 2003; Stavins, 1999), no consensus has been reached regarding the appropriate strategies that can achieve sequestration through economic incentives. Previous studies propose a wide range of policy instruments for regulating forest carbon sequestration. For example, Sohngen and Mendelsohn (2003) propose a carbon rental system that provides rental payments to standing timber and additional payment to carbon permanently stored in wood products. Moreover, van Kooten et al. (1995) suggest a hybrid policy that combines carbon subsidy for growth and a carbon tax for net emissions at the time of harvest. These two approaches are efficiently equivalent and focus on measuring sequestered or emitted carbon. Other proposed approaches, however, deviate from the accurate measurement of carbon to different extents. Some approaches recommend various subsidy schemes based on various discounting factors to account for such problems as permanence, additionality, and leakage (see Kim et al., 2008; Mason & Plantinga, 2013; Tahvonen & Rautiainen, 2017). Still others suggest that payments should be provided for afforestation or avoiding deforestation, but they do not consider the carbon level stored per unit of land (Plantinga et al., 1999). In contrast to approaches that subsidize the forest sector, Searchinger et al. (2009) consider a form of carbon tax to deter land-use change and greenhouse gas (GHG) emissions due to the expansion of wood-based biofuel production. Furthermore, many studies have examined indirect regulation of carbon sequestration in forests through biofuel and other policies (see, e.g., Favero and Mendelsohn 2014; Kriegler et al. 2013; Rose et al. 2012).

Although forest carbon policies may result in changes in forest management, their effects on net carbon emissions are complex. Policies may affect two types of forestry actions: (1) the extensive margin (i.e., the size of forestland) and (2) the intensive margin (i.e., management activities on forestland). In addition, the net carbon effects are a combination of two opposite externalities: (1) a positive externality due to carbon sequestration through biological growth; and (2) a negative externality due to carbon emissions through harvesting for timber, bioenergy, or land-use change. Interestingly, forest management in these two margins influence the carbon flows very differently. In particular, increasing forestland (a decision at the extensive margin) may simultaneously lead to more carbon sequestration (due to more growth of trees) and more carbon emissions (due to more harvests), whereas extending harvest ages (a decision at the intensive margin) may lead to an increase in carbon sequestration and a decrease or increase in carbon emissions.[1] Because of the complicated interaction between forest management and carbon flows, understanding the costs of alternative policies requires a technique that can comprehensively model the dynamic forest management.

In this study, we examine the cost differences among a group of carbon policies based on a one-region, multi-age class, dynamic partial equilibrium forest model. In this model, a planner of the timber market decides how much to harvest and replant trees during each period given the current

[1]The optimal economic rotation age (maximizing net present value, NPV) is less than the maximum sustainable yield (MSY) age, which maximizes total wood output over time. As harvest ages increase from optimal rotation age, the wood output increases on average; as long as the harvest age surpasses the MSY age, an increase in harvest age leads to a decrease in harvest (carbon emissions).

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forest stocks in the region by maximizing the discounted NPV of social welfare derived from the forest sector plus the carbon benefits/costs given the carbon policy imposed on the forest sector. In contrast to the single land class model, this model can capture the dynamic adjustments at the intensive and extensive margins of forest management, which are made in response to carbon policies. At the intensive margin, the harvest area at each age is a continuous choice that can vary over time and across age classes. At the extensive margin, the model replants less or more than the total harvested area but faces the opportunity cost of alternative land use. Six policies are considered: a reference scenario, a scenario that only taxes emissions at the time of harvests, three different per-ton subsidy scenarios, and a per-hectare subsidy scenario.

A particular concern arises with the policy that only taxes carbon emissions from the forest sector. Our analysis indicates that a carbon tax on forest emissions may result in net carbon emissions. The tax-only policy fails to promote carbon sequestration effectively at the intensive margin, because the effects on extending rotation age are very small or even negligible, which is consistent with Gamponia and Mendelsohn (1987). Moreover, the tax-only policy leads to losses of forestland, as the tax on carbon reduces expected land value in forests compared to other uses.[2] The results for the other policy scenarios show that both the per-hectare and per-ton policies provide incentives for afforestation and avoiding deforestation. However, the per-hectare policy is significantly more costly than the per-ton policies,[3] because it is incapable of providing incentives to increase carbon stocks at the intensive margin. For per-ton policies, the cost of carbon sequestration increases by three to five times if forest management is not allowed to be adjusted. For a per-hectare policy, the cost of an additional ton of carbon uptake can range from $98 per MtC to more than $1643 per MtC depending on the level of mitigation target, which is 2–16 times more expensive than the first best per-ton policies. Several robustness checks are conducted to assess the validity of these results for key assumptions on woody bioenergy production, carbon prices, land-use conditions, and yield functions.

This study makes several contributions to the literature. First, the study offers some insights into the cost estimation of carbon sequestration. Although a number of studies has described the factors affecting costs of forest carbon sequestration, such as permanence (Maréchal & Hecq, 2006; Murray et al., 2007), additionality (Mason & Plantinga, 2013), and leakage (Gan & McCarl, 2007; Murray et al., 2004), this study provides evidence that the policy choice could make a big difference in the costs of carbon sequestration. The cost of forest sequestration in the past literature ranges from $10 to $150/MtC (see Richards & Stokes, 2004). Our analysis suggests that the marginal costs could be substantially larger (from $30/MtC to more than $1643/MtC) than the existing literature suggests if the policy proposed does not create incentives to sequester more carbon at the intensive margin. Most studies on carbon sequestration have focused largely on the changes at the extensive margin, that is, afforestation and reforestation (e.g., Lubowski et al., 2006; Plantinga et al., 1999; Stavins, 1999). One example focusing on the role of management activities is Kim and Langpap (2015), who compare carbon sequestration-based and practice-based incentive payment schemes. Cunha-e-Sá et al. (2013) compare several carbon policies in a dynamic forest sector model, but they assume a forest age range much younger than the harvest ages of most managed forests in temperate and tropical regions (Im et al., 2007; van Kooten et al., 1995).[4] As a result, the adjustment at the intensive margin is rather limited. Antle et al. (2003) compare carbon policies for cropland and show that the costs of per-ton and per-hectare programs ultimately converge. Their results rely on heterogeneity across croplands; their intensive margin choices are also limited. By allowing continuous

[2]Note that an implicit assumption we make is that carbon policies in the other land-use sectors remain unchanged. Thus, the changes of carbon policies in non-forest sectors, such as the agricultural sector, are not considered here.

[3]The cost of carbon sequestration is calculated as the welfare losses from the reference scenario (business-as-usual), which is the opportunity cost of carbon sequestration.

[4]Cunha-e-Sá et al. (2013) use different names for per-ton policy and per-hectare policy. The per-ton policy is named the carbon flow approach, and the per-hectare policy is named the average approach. Although they do not provide cost estimates, they show that a per-hectare policy bears welfare losses ranging from 0.42% to 5.56% compared with a per-ton policy. As a robustness check, we test the yield function used by Cunha-e-Sá et al. (2013) in the robustness checks.

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choices in forest management, we show that decisions at the intensive margin contribute to a significant part of differences in efficiency between per-ton and per-hectare policies, and moreover, the efficiency losses from a per-hectare policy increase rather than diminish at a high carbon price.

Second, our study provides an important opportunity to advance understanding of the tax-only policy.[5] Given the recent recommendation by many economists that such countries as the United States should adopt carbon taxes (see *Wall Street Journal*, 2019), it is useful to examine how such a policy could affect forested ecosystems. Although it is correct to tax carbon emissions from energy use directly, our analysis demonstrates that carbon policymakers should treat the land-use sector differently from the energy sector. In particular, our finding sheds light on carbon policy design with woody biomass (e.g., Lintunen & Uusivuori, 2016). Concerns have been raised that increased demand for woody biomass could lead to more carbon emissions owing to land-use changes in the short run. Because of this, some studies and the Manomet Center for Conservation Sciences (2010) suggest that emissions from harvesting wood for bioenergy should be taxed. Such a prescription, although it reduces carbon emissions from burning wood for bioenergy, ignores the contribution of forest management and the associated carbon sequestration with forest growth.

Third, this paper distinguishes from a large body of studies on forest policies using single land class models. The forestry economic literature has a long history exploring the effects of various policies on forest management, including the issue of taxation (e.g., Englin & Klan, 1990; Gamponia & Mendelsohn, 1987) and carbon policies (e.g., Cunha-e-Sá et al., 2013; Im et al., 2007; van Kooten et al., 1995). Traditionally, these issues are examined by using a single land class model in which forests are managed as even aged stands.[6] The models usually assume that timber price and forestland are exogenous. Compared with a single age class model, the partial equilibrium multi-age class model that we develop have several advantages: (1) It allows annual harvests so the model can demonstrate the paths of adjustments of forest stocks prior to the new steady state; (2) in addition to the adjustments of harvest ages, the model can capture policy-induced land-use changes by introducing a rising opportunity cost of alternative land use; (3) we include policy-induced market effects, which are derived from the systematic changes of timber supply at the aggregate level due to carbon policies. A carbon policy may lead to higher or lower timber supply and, thus, indirectly affects the timber price. Failure to account for the market effects would result in over- or underestimation of the efficiencies of carbon policies.

The rest of the paper is organized as follows. The next section develops a stylized optimal control model of the timber market, which allows for dynamic decision making. The third section introduces carbon accounting methods and different carbon policies. The fourth reports the results of numerical Analysis. The last section concludes.

2 | A DYNAMIC OPTIMIZATION MODEL OF THE TIMBER MARKET

The forestry economic literature has a long history exploring the effects of various policies on forest management, including the issue of taxation (e.g., Englin & Klan, 1990; Gamponia & Mendelsohn, 1987) and carbon policies (e.g., Cunha-e-Sá et al., 2013; Im et al., 2007; van Kooten et al., 1995). Traditionally, these issues are examined using a single land class model where forests are managed as even aged stands. The models usually assume that timber price and forestland are exogenous. Following the Faustmann formula, the model maximizes the present value of the taxed income from bare land:

[5]Previous studies focused only on either carbon subsidy schemes (Cho et al., 2018; Cunha-e-Sá et al., 2013) or subsidy and tax schemes (Lintunen & Uusivuori, 2016; van Kooten et al., 1995).

[6]For example, Gamponia and Mendelsohn (1987) and Englin and Klan (1990) employ the Faustmann model to measure the effects of a yield tax. van Kooten et al. (1995) compares the Faustmann optimal rotation age and Hartman optimal rotation age under a carbon tax and subsidy policy.

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$$Max_a \frac{(Pv(a)+f^c(a)-\psi)\beta^a}{1-\beta^a} \tag{1}$$

where P is the exogenous market price of timber products; $v(a)$ measures the volume of timber standing on an hectare of land whose tree are all a years old. f^c is the carbon benefit/cost given the specific carbon policy. ψ is the regeneration cost per hectare, and β is the discount rate. Solving Equation 1 yields a steady-state optimal rotation age (a^*) in response to each carbon policy, which satisfies the following first-order condition:

$$\frac{Pv'(a^*)+f^{c\prime}(a^*)}{Pv(a^*)+f^c(a^*)-\psi} = -\frac{ln\beta}{1-\beta^{a^*}} \tag{2}$$

The single age class model allows only static comparisons of steady-state rotations and fails to take into account the intertemporal policy effects prior to the new steady state, changes at the extensive margin (i.e., land-use changes) in response to alternative policies, and the policy-induced market effects. To address those limitations a single land class model, we present a one-region, multi-age class, dynamic partial equilibrium forest model based on Sohngen and Sedjo (1998). We extend Sohngen and Sedjo (1998) by introducing carbon policies. The model assumes that society may have a certain distribution of trees in the initial period. During each period, a planner of the timber market decides how much to harvest for each age classes and how much to replant. One advantage of the multi-age class model is that it allows annual harvests so the model can demonstrate the paths of adjustments of forest stocks prior to the new steady state. In addition to the adjustments of rotation ages, the model can capture policy-induced, land-use changes by introducing a rising opportunity cost of alternative land use. The objective function for the model is the discounted NPV of the time stream of net surplus (consumers' plus producers' surplus) resulting from the forest sector plus the carbon benefits/costs given the carbon policy imposed on the forest sector. The planner of the timber market decides how much to harvest and replant during each period given the initial forest stocks in the region. Thus, the problem of the planner of the timber market can be expressed as

$$Max_{h_{a,t},g_t} \sum_{t=0}^{\infty} \beta^t \left\{ \int_0^{q_t} D(u)du - \psi g_t - \int_0^{l_t} r(u)du + f_t^c(h_{a,t},x_{a,t}) \right\} \tag{3}$$

subject to

$$x_{a,t+1} = x_{a-1,t} - h_{a-1,t} \text{ for } a>1 \text{ and } t>0 \tag{4}$$

$$x_{1,t+1} = g_t \text{ for } t>0 \tag{5}$$

$$q_t = \sum_a h_{a,t} v(a) \tag{6}$$

$$l_t = \sum_a x_{a,t} \tag{7}$$

$$x_{a,0} \text{ is given}$$

where $h_{a,t}$ is the harvested area for age class a at time t. g_t represents the area of regeneration at time t. β is the annual discount rate. q_t is the harvested volume at time t. $D(\cdot)$ is the demand function for

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wood products, which is decreasing and convex in q_t[7]; ψ is the regeneration cost per unit of land; $r(\cdot)$ represents the unit land rental cost, which is increasing and convex in the total land area l_t (opportunity cost); $f_t^c(h_{a,t}, x_{a,t})$ is the carbon benefit/cost given the carbon policy imposed on the forest sector; $x_{a,t}$ is the current area of forests for age class a at t; $x_{a,0}$ is the current area of forests for age class a in the initial period; and $v(a)$ is the yield function, which measures the standing volume of trees per unit of land.

The optimal condition for $h_{a,t}$ is

$$D(q_t)v(a) + \frac{df_t^c}{dh_a, t} = \beta\left(D(q_{t+1})v(a+1) - r(l_{t+1}) + \frac{df_{t+1}^c}{dh_{a+1,t+1}} + \frac{df_{t+1}^c}{dx_{a+1,t+1}}\right) \tag{8}$$

Equation 8 must be satisfied at all times and for all age classes for harvesting to occur. The optimal condition of harvests indicates that the harvest occurs when the sum of the marginal benefit of harvesting an additional unit of timber of age a at time t is equivalent to the sum of the marginal benefit of harvesting this unit of timber at time $t+1$ (this unit of tree is aged $a+1$ at time $t+1$). The left-hand side (LHS) represents the marginal benefit of harvesting an additional unit of timber of age a at time t and the marginal changes in the carbon benefit at time t due to the harvest of this additional unit. The marginal cost calculation involves three components: (1) the marginal benefit from revenues of harvesting this unit of timber at $t+1$; (2) the opportunity cost of holding an additional unit of forestland from postponing the harvest for another period; and (3) the marginal carbon benefit/cost of having an additional unit of tree stocks of age a + 1 at time $t+1$ and the marginal carbon benefit/cost of harvesting an additional unit of timber of age a at time $t+1$.

The following equation characterizes the optimal condition for the regeneration decision (g_t):

$$\beta^a D(q_{t+a})v(a) + \sum_{u=1}^{a} \beta^u \frac{df_{t+u}^c}{dx_{a+1,t+u}} = \psi + \sum_{u=1}^{a} \beta^u r(l_t) \tag{9}$$

The LHS of Equation 9 represents the marginal benefit of replanting an additional unit of trees given the optimal harvest age a, which consists of two parts: (1) the marginal benefit of harvesting an additional unit of timber of age a at time t + a; and (2) the marginal changes in the carbon benefit/costs from time t to t + a due to an additional unit of forestland. The right-hand side (RHS) shows the marginal cost of replanting an additional unit of trees, which are the regeneration costs and the land rental costs of holding this additional unit of forestland.

The dynamic path of optimal decisions derived from Equations 8 and 9 is economically efficient from the society's perspective. In Data S1B, we prove that the first-order conditions for harvest decisions of a private price taker land-owner's problem are the same as the planner's problem.[8]

3 | CARBON ACCOUNTING AND DIFFERENT CARBON POLICIES

3.1 | Carbon accounting

In this section, we discuss how we account for carbon flows in forests. By sequestering CO_2 and other GHG from the atmosphere, forests can generate carbon credits and reduce the total net

[7]We have a single aggregate demand function for wood, and we do not model heterogeneous products. Thus, we assume that our wood products include wood for both energy production and industrial wood. We include a discussion on the carbon savings from wood bioenergy in the robustness checks.

[8]Although we model a social planner's problem where we maximize the NPV of the net surplus of the timber sector, an alternative objective function commonly used in forestry models is to maximize the NPV of the net income stream of all landowners. This is not an appropriate objective for our study, as we analyze the social cost of carbon sequestration of the forest sector. The use of this objective function would yield a monopoly's solution, which would not be optimal from the society's viewpoint. In Data S1B and C, we also show that the first-order conditions are not the same for a private landowner's problem and the planner's problem. A similar discussion is found in Sedjo and Lyon (1996).

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emissions of the society. Meanwhile, harvesting forests for wood products and land-use changes can also generate additional GHG emissions to the atmosphere. A thorough measurement of the mitigation contribution of the forest sector should account for carbon flows in both directions. Our carbon accounting method is consistent with the carbon flow regime, developed in van Kooten et al. (1995), which measures the total carbon benefits of the forest sector based on annual change in forest carbon stock plus the amount of carbon permanently stored in timber products and landfills. Detailed descriptions of the key components of our carbon accounting methods are found in Table 1.

The net carbon sequestration of the studied forests (c_t^n) at time t is the carbon sequestration during growth (c_t^g) minus the carbon emissions from harvests and processing of wood products $\left(c_t^e\right)$. A net increase in c_t^n means that additional carbon has been removed from the atmosphere; and a decrease in c_t^n means that less carbon has been removed from the atmosphere or additional carbon has been emitted from the forest sector. Therefore, we have

$$c_t^n = c_t^g - c_t^e \tag{10}$$

The carbon sequestration during growth (c_t^g) is calculated as the changes of carbon in standing stocks during two periods, which is a function of the annual growth of trees. This is

$$c_t^g = \phi \sum_a x_{a,t} \dot{v}(a) \tag{11}$$

and

$$\dot{v}(a) = v(a+1) - v(a) \tag{12}$$

where ϕ account for the carbon stocks in dead trees, branches, leaves, understory, and forest floor besides the standing stocks for per-unit merchantable wood.

The carbon emissions from harvests and processing of wood products (c_t^e) are calculated as the difference between carbon stocks in standing stocks (including merchantable wood plus dead trees, branches, leaves, understory, and forest floor) minus the permanent storage of carbon stocks in wood products. That is

TABLE 1 Key factors in carbon accounting

Terminologies relating to carbon flux	Components	Formulas
Carbon sequestration from growths (c_t^g)	Changes in carbon contents in the standing stocks, including merchantable wood, dead trees, branches, leaves, understory, and forest floor	$c_t^g = \phi \sum_a x_{a,t} \dot{v}(a)$
Carbon emissions from harvesting and land use changes (c_t^e)	The difference between carbon stocks in standing stocks (including merchantable wood plus dead trees, branches, leaves, understory, and forest floor) minus the permanent storage of carbon stocks in merchantable wood products.	$c_t^e = \phi \sum_a h_{a,t} v(a) - \eta\delta \sum_a h_{a,t} v(a)$
Net carbon sequestration (c_t^n)	Carbon sequestration from growth minus carbon emissions from harvesting and land use changes	$c_t^n = c_t^g - c_t^e$
Carbon gains (losses) of policy i	The differences of net carbon sequestration (c_t^n) between the policy i and the reference scenario with no carbon policy	
Forest carbon stocks (c_t^s)	Carbon contents in the standing stocks, including merchantable wood, dead trees, branches, leaves, understory, and forest floor	$c_t^s = \phi \sum_a x_{a,t} v(a)$

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$$c_t^e = \phi \sum_a h_{a,t} v(a) - \eta\delta \sum_a h_{a,t} v(a) \tag{13}$$

where η is an estimate of the proportion of "permanent" carbon storage, which is the immediate storage in products minus the present value of losses from turnover as products are used and wood in products decomposes (the pickle rate). δ accounts for the carbon stocks in per-cubic meter of harvested merchantable wood and $\delta < \phi$. The value for relevant parameters ϕ, η, and δ can be found in Table 2.[9]

This carbon accounting method accounts for the carbon contents not only in merchantable wood but also in other components of the standing stocks, including residues and soil carbon. When the trees are harvested or forests are shifted to other uses, the model assumes that carbon from those sources is emitted immediately except for the part that is permanently stored in merchantable wood products.[10] When the forests are harvested and replanted, we assume the same accounting of emissions as clear cut; however, after the harvests, the forest carbon accumulates on the site according to the yield function starting from age 0.

The forest carbon stocks, which are a function of the total forest stocks, include carbon contents in merchantable wood, dead trees, branches, leaves, understory, and forest floor. Therefore, the accounting formula is

$$c_t^s = \phi \sum_a x_{a,t} v(a) \tag{14}$$

3.2 | Policy scenarios and optimal conditions

The payments of the five carbon policies we examine are calculated based on the abovementioned carbon accounting. We consider six policy scenarios. These scenarios can be divided into three general groups: (1) a reference scenario in which no carbon policy is applied; (2) two first-best policies in which credits are given by measuring the carbon flow through forests: carbon subsidy and tax and carbon rental; and (3) three non-first-best policies in which only parts of the carbon flow are measured or the carbon flow is not measured at all: carbon tax, carbon subsidy, and land subsidy. The carbon gains (or losses) of each policy are measured as the differences of net carbon sequestration (c_t^n) between the studied policy and the reference scenario with no carbon policy. In the reference scenario, there exists no carbon policy, and the landowners receive revenue only from the timber market.

The two first best policy scenarios are specified as follows:

1. Carbon tax and subsidy (subsidy and tax): This policy, proposed by van Kooten et al. (1995), is a combination of carbon tax and carbon subsidy policies. Carbon sequestered each year through forest growth is subsidized at the current carbon price, and carbon emissions are taxed at the carbon price upon harvest.
2. Carbon rental (C rental): This policy proposes another approach to solve the impermanence of carbon storage in forests by treating the carbon service as temporary. To capture the timing of

[9]We assume that 35% of carbon in merchantable wood has been permanently stored (Daigneault et al., 2010). If we account for all the carbon stocks with standing stocks and other components of forest sites, this is about 18% of the total forest carbon sink. Earles et al. (2012) also show that for Europe, the United States, and Canada the fraction is around 36%, whereas that for the rest of world ranges from 5% to 65%.

[10]We assume that emissions of everything else not stored occurs immediately at harvest. An ideal way of accounting is to let the emissions occur over time and to treat some of the remainder as an additional stock. We do not choose this option owing to lack of parameters and too many complications.

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TABLE 2 Key parameters and sources

Parameters	Descriptions	Value	Source
m	Yield fucntion parameter	7.82	Sohngen & Sedjo, 1998
n	Yield function parameter	52.9	Sohngen & Sedjo, 1998
σ	Price elasticity of demand for wood products	−0.5	Simangunsong and Buongiorno (2001)
ε	Price elasticity of land supply	0.33	Lubowski et al., 2006
ϕ	Carbon conversion rate in standing stocks, dead trees, branches, leaves, understory, and forest floor	0.38 tC/m^3	Tian et al. (2018)
δ	Carbon conversion rate in merchantable wood	0.20 tC/m^3	van Kooten et al. (1995)
η	The pickle rate of wood products	0.35	Daigneault et al. (2010)
β	Discount rate	0.95	
ψ	Regeneration cost	$300 per hectare	Gamponia & Mendelsohn, 1987

carbon flows accurately, this policy measures carbon storage per year. Rather than using the carbon price, p_t^c, for temporary storage of carbon, the payment relies on the rental rate of carbon r_t^c, which is the value of storing carbon for a year. r_t^c is as follows

$$r_t^c = p_t^c - \beta p_{t+1}^c \tag{15}$$

Under the carbon rental system, the forest sector is rewarded each year for the current forest carbon stocks at the rate of r_t^c and for the carbon stored permanently in wood products at the rate of p_t^c.

Table 3 reports a summary of the payment formula of different carbon policies. Table 4 show the optimal conditions for harvest and regeneration decisions under the two first best policies, respectively. One direct finding generated from Table 4 is that the optimal conditions generated under subsidy and tax and C rental are equivalent although their payments differ from each other.[11] This indicates that both subsidy and tax and C rental policy are first-best policies, because they are capable of accurately measuring the carbon flux with respect to forests.

The three non-first-best policies are specified as follows:

1. Carbon subsidy (subsidy): This policy only subsidizes carbon uptake by measuring growth according to the current carbon price. Carbon emissions are not taxed. Additional marginal benefits occur when the harvests are postponed for another year, because more subsidies are earned from the additional year of growth.
2. Land subsidy (land): Land subsidy is a per-hectare policy that provides a subsidy for each unit of forested land, provided that the land is not replaced for uses other than forests after clear cutting. If forests are harvested and replanted, we treat the land as forested land; if forests are clear cut for other uses, the land is no longer eligible for the land subsidy. The rate of the subsidy (s_t^l) is equivalent to the carbon offsets value per unit of land in the reference scenario.[12] The land subsidy policy provides payments according to the total forest area (l_t).

[11]By rearranging the terms in the first-order conditions with respect to subsidy tax and C rental and combining Equation 15, we obtain the same optimal conditions for harvests and regeneration with respect to subsidy and tax and C rental.

[12]The land carbon subsidy paid to per unit of forestland is the carbon value of the average level of net carbon sequestration in the reference scenario. The carbon price used is consistent with other carbon scenarios. The formula of the land subsidy is thus: $s_t^l = \frac{p^c c_{\text{Base}}^n}{l_{Base}}$, where s_t^l is the subsidy on per unit of land in period t, p^c is the carbon price, c_{Base}^n is the net carbon sequestration in the reference scenario, and l_{Base} is the area of forestland in the reference scenario.

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TABLE 3 Carbon benefits/costs under different policies

Scenarios	Carbon policy	Carbon benefits/costs
First-best policies	Carbon tax and subsidy	$f_t^c = p_t^c c_t^g - p_t^c c_t^e$
	Carbon rental	$f_t^c = r_t^c c_t^s + p_t^c \eta \delta \sum_a h_{a,t} v(a)$
Non-first best policies	Carbon tax	$f_t^c = -p_t^c c_t^e$
	Carbon subsidy	$f_t^c = p_t^c c_t^g$
	Land subsidy	$f_t^c = s_t^l l_t$

Note: The carbon rental rate is calculated according to the following rule: $r_t^c = p_t^c - \beta p_{t+1}^c$ (see Sedjo and Marland 2003); s_t^l is the subsidy rate per unit of land at t.

3. Carbon tax: Only carbon emissions upon harvesting and burning wood for energy are taxed. The proportion of carbon that is permanently stored in wood products is deducted from the tax.[13]

Table 5 shows the optimal conditions for harvest and regeneration decisions under three non-first-best policies, respectively. Because the incentives are designed differently from the first best policies, we can infer that the equilibrium harvests, replants, forest area, and prices should be different from the first best policies. Although the optimal conditions in Tables 4 and 5 show how the carbon benefits/costs may affect the determinants of harvests and regeneration, we cannot solve the dynamic optimal decisions analytically owing to the complexity and non-linearity of the system of equations. We further explore the effects in the numerical analysis in the next section.

4 | NUMERICAL ANALYSIS

4.1 | Model specifications and parameters

Using the model presented in the previous section, we examine the effects of alternative carbon policies on timber management and on the timber market through numerical simulation. We focus on dynamic timber market responses to carbon policy changes. The model records forest areas by age class. The future path of output of wood, land supply, and forest stocks are determined endogenously and simultaneously. Accordingly, results in carbon gains can also be inferred based on timber harvests, regeneration of trees, and forest growth. All results are reported in aggregate numbers, although they are consistent with individual landowners' optimization behavior in a competitive timber market. A detailed description of key parameters of this analysis is provided in Table 2. The timber market planner is subject to a constant elasticity timber demand function:

$$D(q_t) = Aq_t^{1/\sigma} \tag{16}$$

where σ is the price elasticity of demand for wood products. The forest area is adjusted through harvests and regeneration according to a constant elasticity land supply function, which captures the opportunity cost of maintaining land in forests instead of alternative uses:

$$r(l_t) = Rl_t^{1/\varepsilon} \tag{17}$$

[13]One could imagine a more sophisticated tax which is levied on current emissions plus the foregone sequestration compared with the first best policies. It is not difficult to find that the new tax is equivalent to the two first best policies that we proposed. The only differences are the differences in distribution. As a result, the optimal solutions for this sophisticated tax should be the same as the first best policies.

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TABLE 4 Optimal conditions for first best policies

Carbon tax and subsidy	$[h_{a,t}]$	$D(q_t)v(a)-(\phi-\eta\delta)p_t^c v(a)=\beta\left(D(q_{t+1})v(a+1)-r(l_{t+1})-(\phi-\eta\delta)p_{t+1}^c v(a+1)+\phi p_{t+1}^c \dot{v}(a+1)\right)$
	$[g_t]$	$\beta^a\left(D(q_{t+a})-(\phi-\eta\delta)p_t^c\right)v(a)+\phi\sum_{u=1}^{a}\beta^u p_{t+u}^c \dot{v}(u)=\psi+\sum_{u=1}^{a}\beta^u r(l_t)$
Carbon rental	$[h_{a,t}]$	$D(q_t)v(a)+p_t^c\eta\delta v(a)=\beta\left(D(q_{t+1})v(a+1)-r(l_{t+1})+p_{t+1}^c\eta\delta v(a+1)+\phi r_{t+1}^c v(a+1)\right)$
	$[g_t]$	$\beta^a\left(D(q_{t+a})+\eta_t^c\right)v(a)+\phi\sum_{u=1}^{a}\beta^u r_{t+u}^c v(u)=\psi+\sum_{u=1}^{a}\beta^u r(l_t)$

Note: This table shows the optimal conditions for the scenarios of carbon tax, carbon subsidy, and land subsidy, respectively. The optimal conditions in the third column are derived by taking first order conditions with respect to the choice variables in the second column. $h_{a,t}$ represents the area of harvest at age a in period t; g_t represents the area of harvest at age a in period t.

TABLE 5 Optimal conditions for non-first-best policies

Carbon tax	$[h_{a,t}]$	$D(q_t)v(a)-(\phi-\eta\delta)p_t^c v(a)=\beta\left(D(q_{t+1})v(a+1)-(\phi-\eta\delta)p_{t+1}^c v(a+1)-r(l_{t+1})\right)$
	$[g_t]$	$beta^a\left(D(q_{t+a})-(\phi-\eta\delta)p_t^c\right)v(a)=\psi+\sum_{u=1}^{a}\beta^u r(l_t)$
Carbon subsidy	$[h_{a,t}]$	$D(q_t)v(a)=\beta\left(D(q_{t+1})v(a+1)-r(l_{t+1})+\phi p_{t+1}^c \dot{v}(a+1)\right)$
	$[g_t]$	$\beta^a D(q_{t+a})v(a)+\phi\sum_{u=1}^{a}\beta^u p_{t+u}^c \dot{v}(u)=\psi+\sum_{u=1}^{a}\beta^u r(l_t)$
Land subsidy	$[h_{a,t}]$	$D(q_t)v(a)=\beta\left(D(q_{t+1})v(a+1)-r(l_{t+1})+s_{t+1}^l\right)$
	$[g_t]$	$\beta^a D(q_{t+a})v(a)+s_t^l=\psi+\sum_{u=1}^{a}\beta^u r(l_t)$

Note: This table shows the optimal conditions for the scenarios of carbon tax, carbon subsidy, and land subsidy, respectively. The optimal conditions in the third column are derived by taking first-order conditions with respect to the choice variables in the second column. $h_{a,t}$ represents the area of harvest at age a in period t; g_t represents the area of harvest at age a in period t.

where ε is the price elasticity of land supply. Therefore, land might be more expensive to rent if new land is added to forests. The forest stocks modeled are plantations of US southern pine, which is a representative species for timber production in the United States. The yield function follows the following form (see Figure 11), where m and n are parameters:

$$v(a)=exp\left(m-\frac{n}{a}\right) \tag{18}$$

The carbon prices in the main analysis are constant over time, ranging from \$10 to \$400 per MtC (equivalent to \$2.7 to \$108.9 per ton of CO_2, respectively). The results for a rising carbon price are presented in the sensitivity analysis. The model is solved using the non-linear optimization solver of the general algebraic modeling system, with 1-year steps for a total of 300 years. The numerical model generates the paths of the harvest age, timber output, forest area, and carbon gains for all scenarios, with each scenario labeled differently. In the reference scenario, the optimal output, harvests, and inventory paths over the projection period remain unchanged. Thus, the optimal harvest age in the reference scenario is chosen as 28 years, and the replant area is equivalent to the harvest area during each period. Carbon uptake per year does not vary, because the annual growth is equivalent only to the amount of timber harvested from the forests. We compare the effects of each policy scenario with those of the reference scenario.

4.2 | The dynamic paths

4.2.1 | First-best policies

Figure 1 reports the dynamic paths of average harvest age (a), forest area (b), output wood products (c), and net carbon sequestration (d), respectively, at a carbon price of \$40 per MtC. Carbon subsidy

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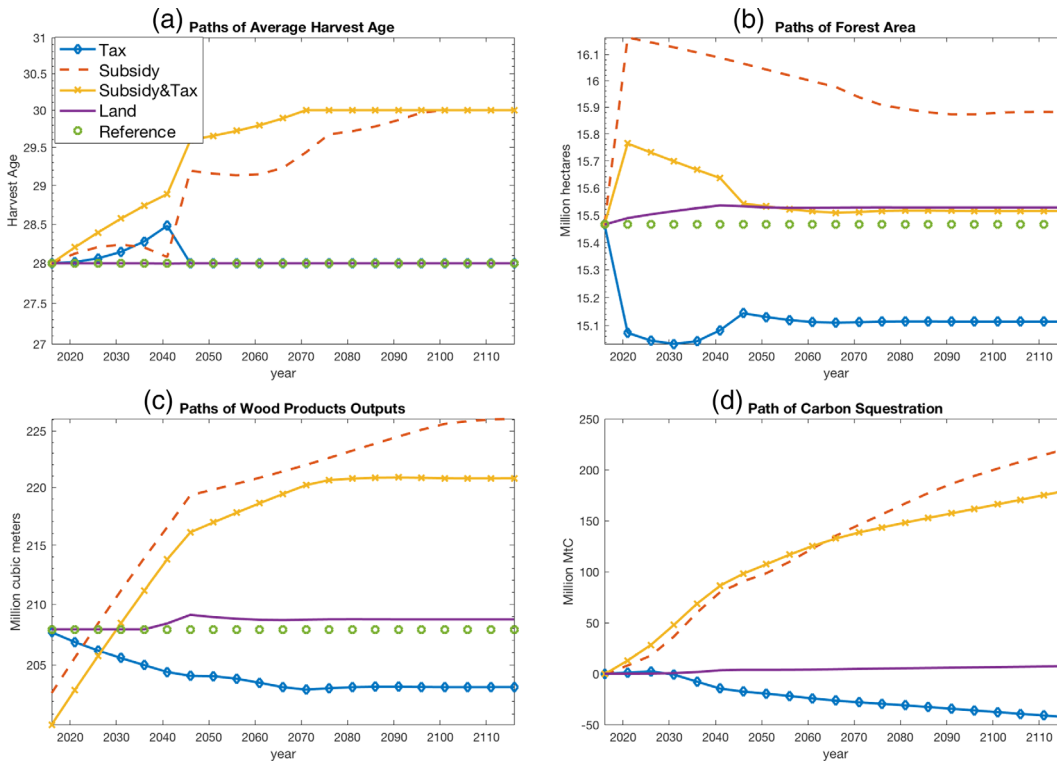

FIGURE 1 Dynamic paths-constant C prices
Notes: $p_t^c = \$40/\text{MtC}$; yield function: US southern pine; $\sigma = -0.5$; $\varepsilon = 0.33$; the results of subsidy and tax and C rental are identical so we combine their results

and tax and C rental create the same marginal incentives for managing forests and sequestrating carbon. In the simulation, the two scenarios exert identical effects on management practices, forest area, and wood output, and thus, we combine their results. Both policies cause the harvest age to rise to the same level as under the carbon subsidy scenario (Figure 1a). The total forest area increases to a lesser extent under the subsidy and tax scenario than under the subsidy-only scenario, because the tax element reduces the value of the land (Figure 1b). The trend of the carbon offset effects is similar to that of the carbon subsidy scenario. More carbon offsets are achieved in the forest sector relative to the reference scenario (Figure 1d).

4.2.2 | Carbon subsidy

The introduction of the carbon subsidy extends the harvest age. The average harvest age increases from 28 years initially to 30 years by 2110. The forest area expands from 15.46 million hectares to 15.88 million hectares by the end of the century. This is consistent with the expected result. The subsidy provides additional benefits to the forest sector and, thus, encourages afforestation through land conversion from other sectors. The wood output under the carbon subsidy policy decreases relative to that of the reference scenario initially and then gradually increases and surpasses that of the reference scenario. This observation is due to the perfect foresight of the optimal control model. When the planner of the timber market foresees a carbon subsidy, it adjusts management to increase the annual growth of trees at the extensive and intensive margins. At the extensive margin, the planner increases the forest area. At the intensive margin, the planner needs to shift harvest ages to older age

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classes. Thus, in the beginning period, the planner holds part of the stocks from harvest until the stocks are older, because the trees take time to mature. This explains the lower initial level of output relative to the reference scenario. Due to changes in the intensive and extensive margins, the carbon subsidy positively affects net carbon sequestration.

4.2.3 | Land subsidy

As for the land subsidy, the most important effects on management practices are the positive effects on the forest area (Figure 1a). The total forest area increases over time. Regarding the effects on harvest ages, the results from the Faustmann model state that lump-sum transfers by forestland (e.g., the lump-sum tax illustrated by Koskela and Ollikainen, 2001) provide no incentive for extending harvest ages, because these transfers do not create any incentive for management on the stands. Consistently, we find that the per-hectare policy causes zero changes in average harvest age. The carbon effects are positive and significant. However, the carbon gains from the land subsidy are much smaller (less than 5%) than those under the subsidy and tax (C rental) scenarios. The gap cannot be fully explained by the land expansion schedules of these scenarios, because the rates of the land expansion of other scenarios are approximately the same as that of the land subsidy scenario. The difference in total carbon offsets is mostly attributed to their variances at the intensive margin. The land subsidy provides few economic incentives at the intensive margin; therefore, its potential to promote carbon sequestration by changing management practices is much lower than that of per-ton policies. Consequently, the effectiveness of the land subsidy is highly diminished. We further demonstrate the efficiency losses from the land subsidy by comparing their respective marginal cost curves.

4.3 | Carbon tax

Figures 2–4 report the simulation results under the carbon tax scenario. The dynamic system reaches a steady state about three decades after the introduction of the carbon tax. At the extensive margin, the forestland area decreases (Figure 2b) as more forestland is clear cut than replanted in the first decade after the tax is introduced. As a result, at the extensive margin, the tax causes more forestland to be used for alternative purposes, which reduces forest inventories as a whole.

An interesting finding arising from Figure 2a is that the harvest age at the steady state remains the same as the level in the reference scenario. This posits a potential inconsistency between our

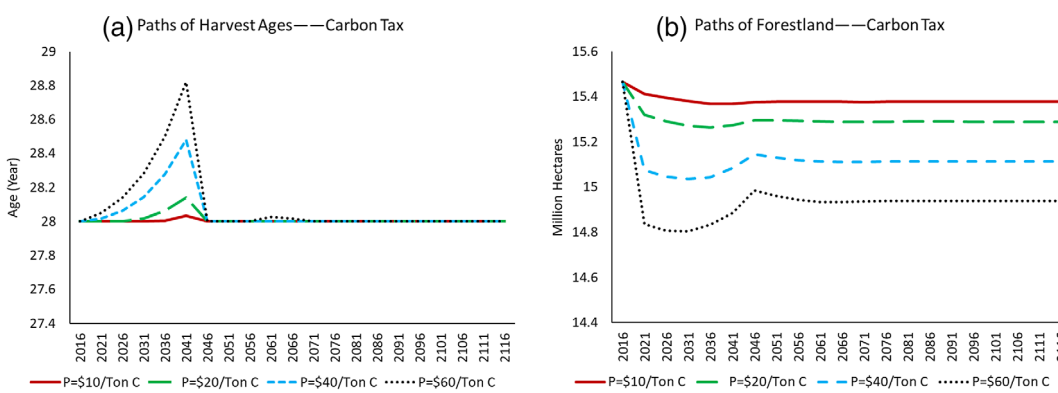

FIGURE 2 Paths of harvest ages and forestland of carbon tax under alternative carbon prices
Notes: C price: Constant C prices ($10 to $60/MtC); yield function: US southern pine; $\sigma = -0.5$; $\varepsilon = 0.33$

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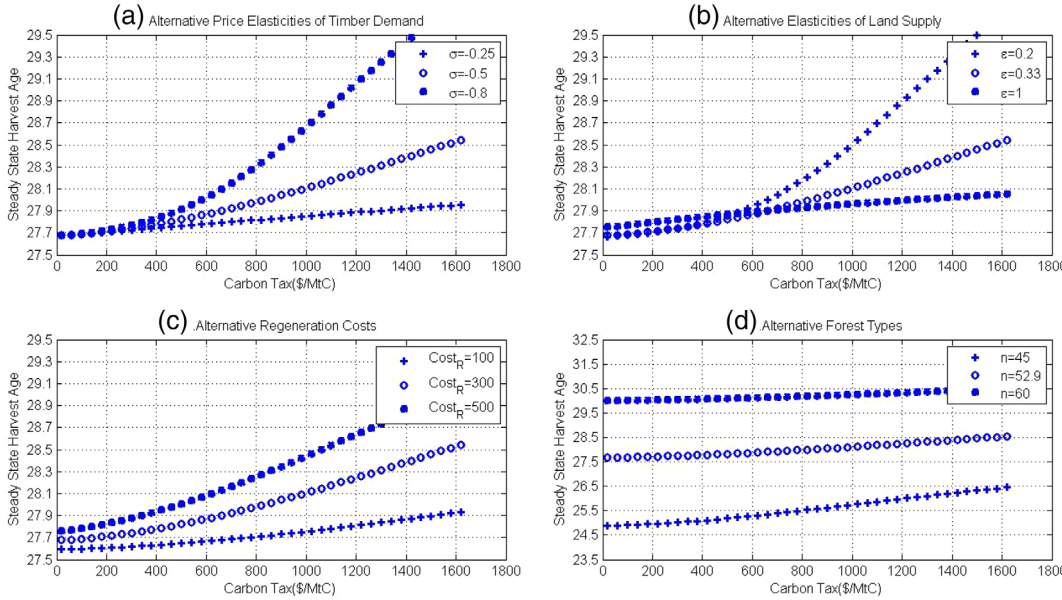

FIGURE 3 Effects of carbon tax on steady state harvest ages

Notes: This figure shows the continuous effects of a carbon tax on harvest ages. Column a–c represent alternative assumptions of price elasticities of timber demand, elasticities of land supply, regeneration costs, and yield functions, respectively

analysis and the existing literature. Englin and Klan (1990) and Gamponia and Mendelsohn (1987) find that a yield tax, which is equivalent to a carbon tax, has a positive effect on rotation ages. The explanation for this inconsistency is that the positive effects are quantitatively very small. Note that the dynamic optimization timber market model we construct is a discrete model. The time step of the model is 1 year so that the harvest occurs once a year. As a result, the model cannot identify changes in harvesting time smaller than 1 year. In fact, Gamponia and Mendelsohn (1987) show that even with a 40% yield tax, the change in rotation ages is less than 1 year.

In order to further verify the small effect of a yield/carbon tax on rotation age, we solve the effects of the carbon tax on optimal harvest age a^* in the steady state by assuming that the harvest age is a continuous variable.[14] The results are reported in Figure 3. Figures 3a–d display the results under alternative assumptions of price elasticities of timber demand, elasticities of land supply, regeneration costs, and forest types, respectively. In all cases, the carbon tax has a positive effect on harvest ages. More importantly, the scale of effects is very small. Under the base case assumptions (middle line), with a carbon price from \$0–\$1500/MtC (equivalent to a 0–380% yield tax), the steady-state rotation changes ranges from 27.6 to 28.5 years, implying that to achieve an increase in harvest age of 1 year, the carbon price has to be as high as \$1500/MtC. The scale of effects becomes larger with a higher price elasticity of timber demand, a lower elasticity of land supply, or a higher regeneration cost. However, the harvest ages remain below 28.5 years under carbon prices lower than \$1000/MtC in all cases.

Figure 4 compares the dynamic paths of harvest ages, forest area, timber prices, timber outputs, and carbon sequestration in the base case with those in alternative demand and land supply scenarios. In all cases, there is little change in harvest ages in the steady state while the forestland decreases. Moreover, the timber outputs reduce and the timber prices rise. Consequently, we observe that the carbon tax scenario leads to higher net carbon emissions beyond the reference scenario. The carbon

[14]We assume that in the steady state, forests are equally distributed across age classes from 1 to optimal rotation age a^*, and the optimal decision is to clear cut all the trees at a^* and to replant the same area ($\frac{F}{a^*}$).

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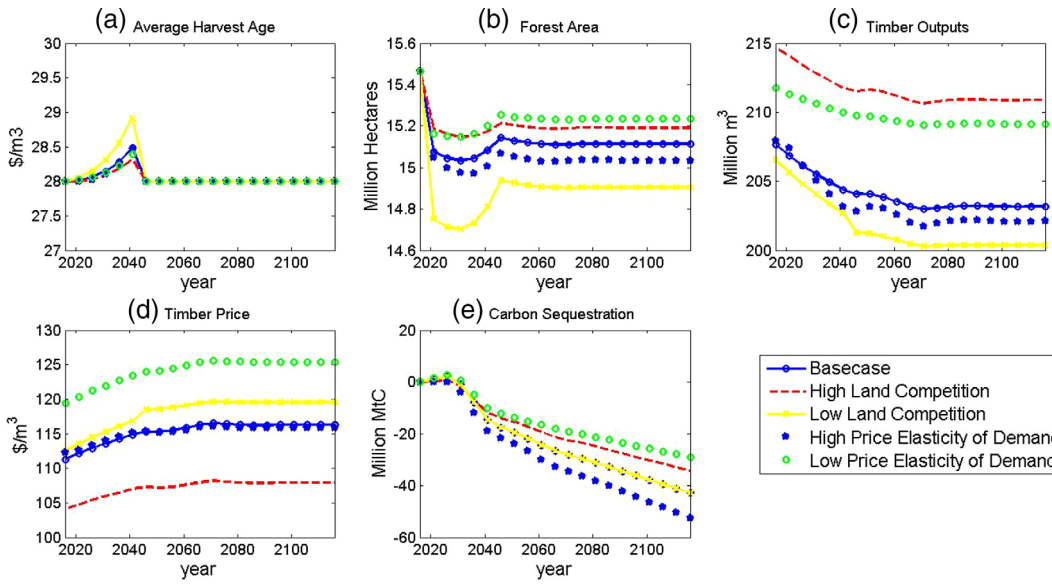

FIGURE 4 Dynamic paths-carbon tax

Notes: $p_t^c = \$40/\text{MtC}$; yield function: US southern pine; base case: $\sigma = -0.5$; $\varepsilon = 0.33$; high land competition: $\sigma = -0.5$; $\varepsilon = 0.2$; low land competition: $\sigma = -0.5$; $\varepsilon = 1$; high price elasticity of demand: $\sigma = -0.91$; $\varepsilon = 0.33$; low price elasticity of demand: $\sigma = -0.25$; $\varepsilon = 0.33$

tax on forests also results in welfare losses in the timber sector beyond the reduction in net carbon sequestration. Figure 5 shows the total costs in welfare losses at each level of net carbon emissions with respect to the reference scenario. The more elastic the land supply function (lower land competition), the higher the total economic cost; similarly, the more elastic the demand function, the higher the total economic cost.

A carbon tax has been proposed for the purpose of reducing carbon emissions by creating a disincentive for emissions due to harvests or other uses of wood (Searchinger et al., 2009). However, the tax also reduces the rate of carbon sequestration so the net carbon effects may be negative. Note that even though we find very small positive effects of the carbon tax on harvest ages in a continuous setting, the general findings on negative carbon effects generated from a carbon tax still hold for two reasons. First, in many places, tree harvesting may be subject to seasonal restrictions for such reasons as maintaining forest health and environmentally protected species. Therefore, those restrictions may limit the harvesting of timber during portions of the year (US Forest Service, Northwest Research Station, 2013). As a result, there may be no noticeable change in the actual time of harvesting a tree in practice. Second, even if private owners respond to the carbon tax by delaying the harvest timing, the carbon losses due to forestland losses still dominate any resulting carbon gains at the intensive margin. According to our simulation in the base case, for example, under a carbon tax of \$800/MtC, the steady-state rotation age changes from 27.6 to 28.5 years, with total forestland contracting from 15.52 million hectares to 7.82 million hectares. The extension of rotation age increases the annual net carbon sequestration on per hectare of land by only 1.81%, whereas the total annual net sequestration reduces by 52%, mainly due to loss of forestland. This result implies that a carbon tax on forest emissions results in net carbon emissions without subsidizing the carbon sequestration from tree growth.

5 | MARGINAL COST CURVES

To derive the marginal costs of carbon sequestration under different policies, we first compute an implicit cost that is equivalent to the losses of surplus in the timber sector at each carbon price. We

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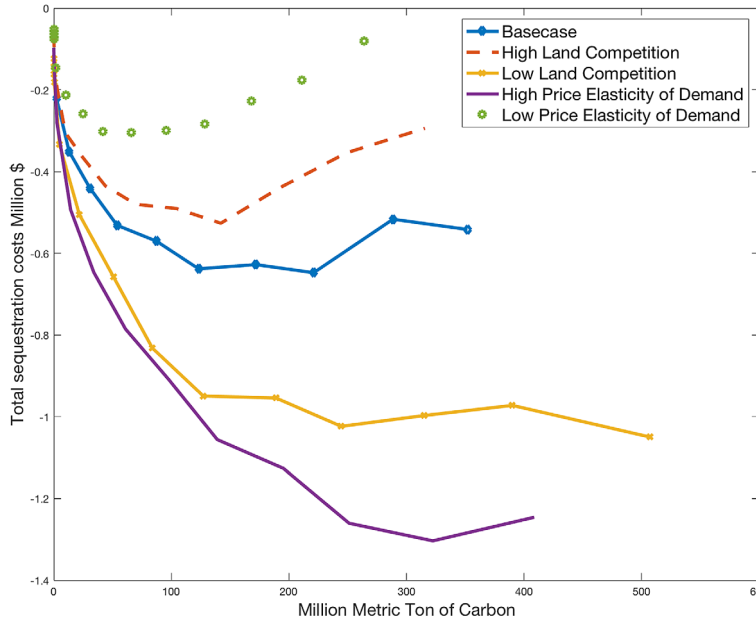

FIGURE 5 Total cost curves of carbon sequestration-carbon tax

Notes: $p_t^c = \$10$ to \$200/MtC; yield function: US southern pine; base case: $\sigma = -0.5$; $\varepsilon = 0.33$; high land competition: $\sigma = -0.5$; $\varepsilon = 0.2$; low land competition: $\sigma = -0.5$; $\varepsilon = 1$; high price elasticity of demand: $\sigma = -0.91$; $\varepsilon = 0.33$; low price elasticity of demand: $\sigma = -0.25$; $\varepsilon = 0.33$

run the timber market model multiple times at different constant carbon prices ranging from \$10 to \$400/MtC (p_n^c, n = 1,2,.....,). At a carbon price of p_n^c, the economic cost of carbon sequestration ($Tcost_n$) is calculated as the differences in NPV of flows of consumers' plus producers' surplus from the timber market for the first 100 years between specific policy scenarios and the reference scenario with no carbon policy. Furthermore, the associated total carbon gain ($Tcarbon_n$) is the sum of discounted differences of net carbon sequestration (c_t^n) between policy i and the reference scenario with no carbon policy. Subsequently, the implicit cost is matched with its level of carbon gain at the given carbon price. Finally, the marginal cost per ton is determined by linking each increment of the implicit cost to the additional carbon sequestration triggered by this specific increment.[15] The marginal costs per ton of carbon sequestered for each type of policy are presented in Figure 6. Note that the marginal cost under the carbon tax scenario is not presented, because it leads to negative carbon gains.

The land subsidy results in higher efficiency losses than other policies. Among all scenarios creating positive carbon offsets, the lowest marginal cost is incurred under the subsidy and tax and C rental policies, as expected. The next efficient policy is the carbon subsidy policy; its marginal cost curve is slightly higher than those of the two optimal policies. The marginal cost of the land subsidy policy is substantially higher than those of the two optimal policies and the carbon subsidy policy. To achieve the same sequestration target, the land subsidy policy is 3–16 times more expensive than other policies. Moreover, the efficiency losses increase with carbon prices. For example, at a carbon price of \$112/MtC (equivalent to \$30/Mt CO_2), the carbon supply under the land subsidy scenario is 10.1% of that under the subsidy and tax scenario and is 14.1% of that under the carbon subsidy scenario. Moreover, at \$224/MtC (equivalent to \$60/Mt CO_2), the carbon supply

[15]In the calculation, we convert to an annual level using returns of the periodic payment for an annuity so that the quantity of marginal cost function reflects an annual carbon sequestration level.

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is only 6.3% of that under the subsidy and tax scenario and is 9.1% of that under the carbon subsidy scenario.

We also compare our results with those of Antle et al. (2003). They find that the marginal cost curves of per-ton and per-hectare policies ultimately converge at a high carbon price. However, the curves of per-ton and per-hectare policies converge only when the carbon policies do not alter the land value and, thus, do not alter the extensive margin or the land area devoted to sequestration. In our model, the intensive and extensive margins are endogenous; thus, per-hectare carbon policies are significantly more inefficient than per-ton carbon policies. Moreover, the inefficiency exists regardless of their effects at the extensive margin. Even if the same forests were included in the model, the effects on harvest ages would still be dramatically different between the two types of policies.

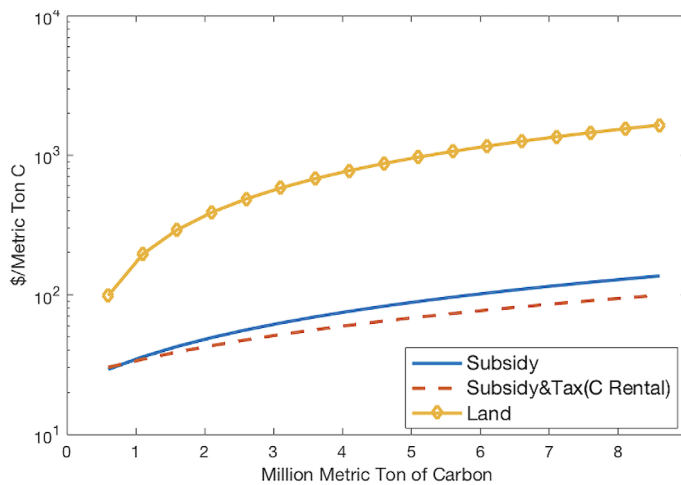

FIGURE 6 Marginal cost curves-base case
Notes: $p_t^c = \$10$ to $ 200/MtC; yield function: US southern pine; $\sigma = -0.5$; $\varepsilon = 0.33$; the results of subsidy and tax and C rental are identical so we combine their results

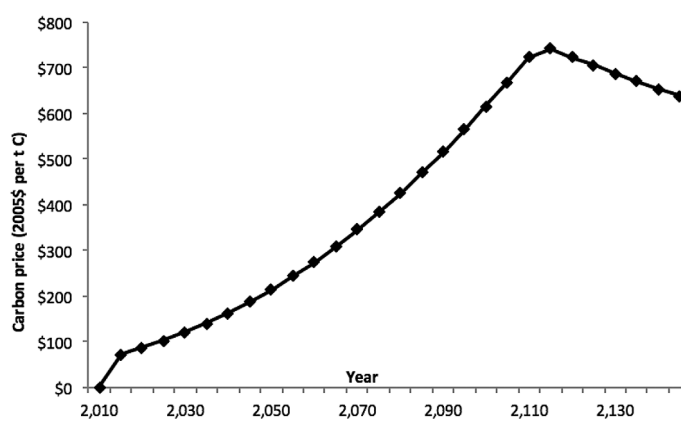

FIGURE 7 Rising carbon price
Notes: The carbon prices are estimated from the optimal carbon tax scenario of DICE-2010 model (Nordhaus, 2010), which represents the social optimal marginal costs of an additional ton of CO_2 emissions to the atmosphere

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6 | ROBUSTNESS CHECKS

We conduct several robustness checks to assess the validity of the results. First, up to now, we have assumed a constant carbon price. It would be interesting to run the model under a different carbon

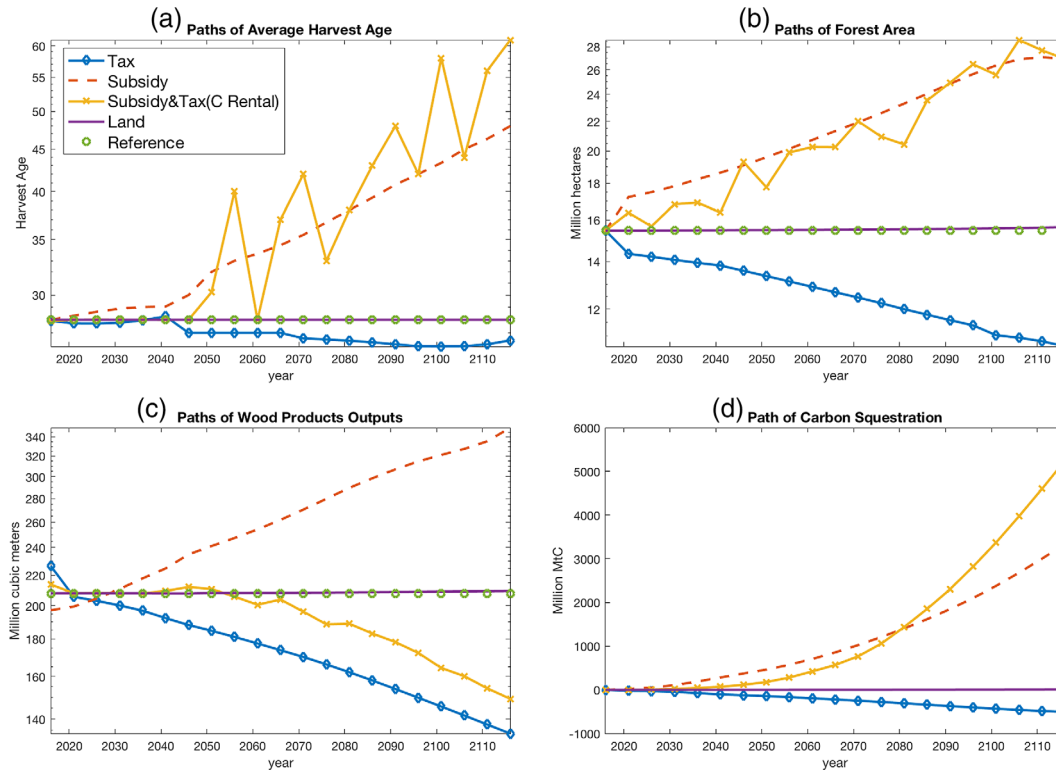

FIGURE 8 Dynamic paths-rising carbon price. C Price: The optimal carbon tax scenario of DICE-2010 model (Figure 7); yield function: US southern pine; σ: −0.5; ε:0.33; the results of Subsidy&tax and C rental are identical so we combine their results

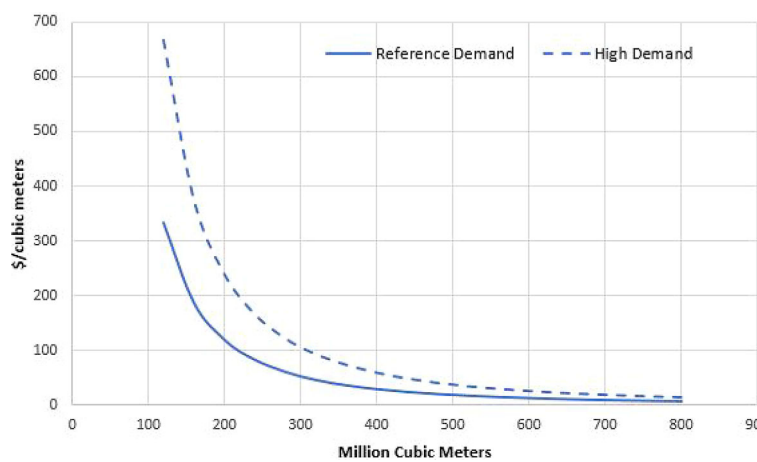

FIGURE 9 Alternative demand functions

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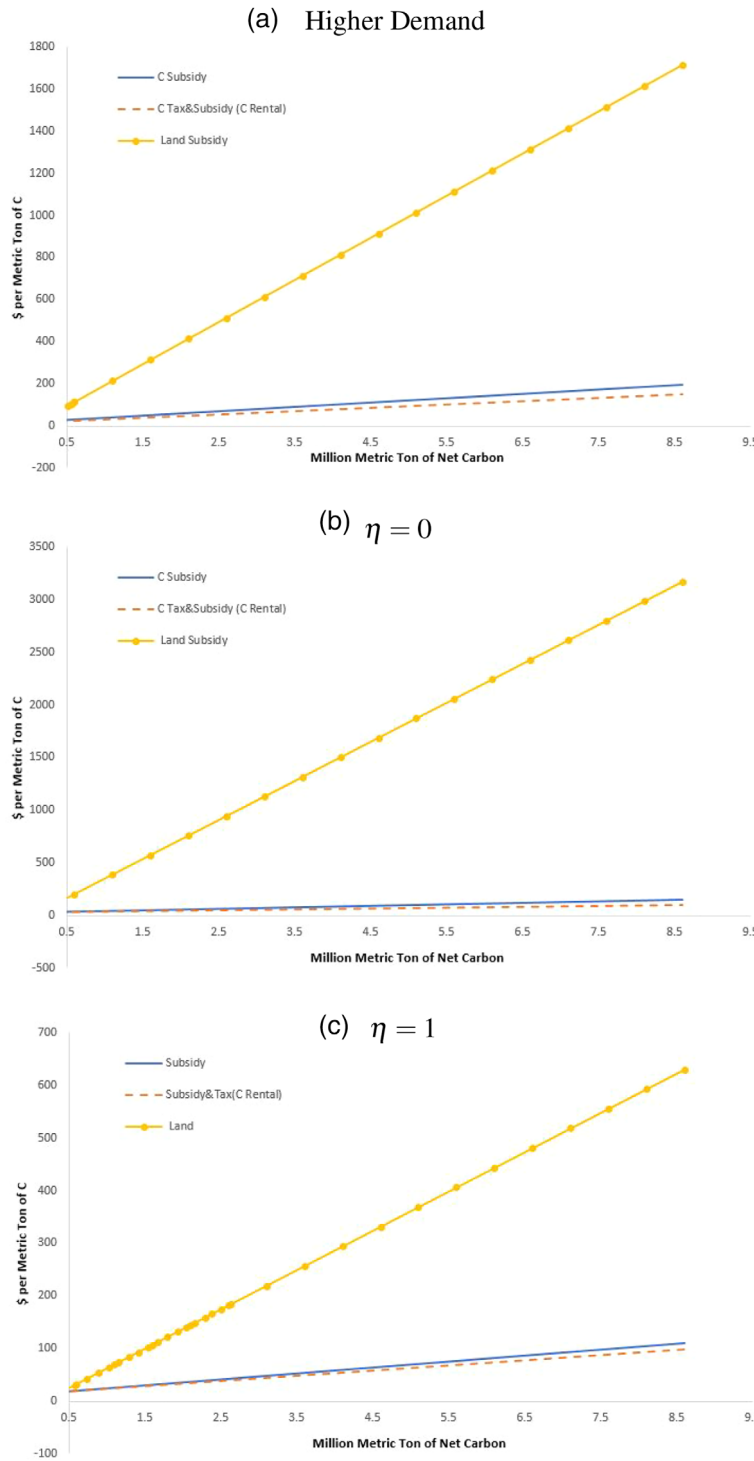

FIGURE 10 Marginal cost curves-woody bioenergy

Notes: Figure 10 Depicts the marginal cost curves under alternative assumptions. (a) Assumes a higher demand function, which is displayed in Figure 9; (b) and (c) assumes the pickle rate η equals 0 and 1, respectively. $p_t^c = \$10$ to \$ 200/MtC; yield function: US southern pine; $\sigma = -0.5$; $\varepsilon = 0.33$; the results of Subsidy&tax and C rental are identical so we combine their results

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price path, because different paths could affect the decisions of forest management. Second, although woody bioenergy plays an increasingly important role in energy supply, our model does not directly include woody bioenergy production. We test whether our main findings still hold when accounting for carbon savings from the use of woody bioenergy for fossil fuel substitution. Third, there are additional concerns about whether the result depends on the assumption of the land-use situation and the yield function. We rule out these possibilities by testing alternative land supply species and tree yield functions.

6.1 | Rising carbon prices

We first test a rising carbon price path. The carbon prices (Figure 7) are estimated from the optimal carbon tax scenario of the DICE-2010 Model (Nordhaus, 2010), which represents the social optimal marginal costs of an additional ton of emissions to the atmosphere. The projection uses the carbon price in 2010 as the starting level, and the price increases to as high as \$700/MtC (equivalent to approximately \$190/ton of CO_2) by 2110. The results are reported in Figure 8. The results are generally consistent with earlier results, but the magnitude of policy effects (e.g., on harvest ages, forest area, and net carbon sequestration) is much larger than the earlier results. The effects also increase over time as the carbon price increases. Contrary to earlier results under constant carbon prices, the harvest ages for the tax-only policy are found to be lower than the level in the reference scenario. The observed decrease may be attributed to the perfect foresight of the carbon price path in the model. When the forest sector foresees that the carbon price is higher in the future, and so too is the tax rate, it harvests more frequently than in the case of a constant carbon price. Meanwhile, more forests are harvested than replanted under the tax-only policy. This combination of findings supports the conclusion that the carbon tax leads to net carbon emissions compared with the reference case under rising carbon prices.

6.2 | Woody bioenergy production

Although the current model does not directly include a pathway for bioenergy, the current framework can implicitly incorporate the woody bioenergy demand and supply. Specifically, the use of

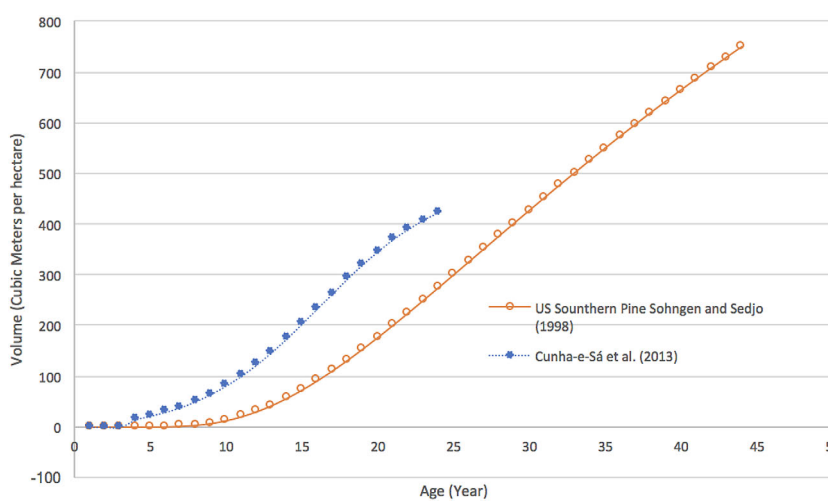

FIGURE 11 Alternative yield functions

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bioenergy may result in changes in carbon accounting in two ways: (1) it alters the path of carbon emissions after the trees are harvested; and (2) it increases the demand for wood products. The changes in net carbon emissions with bioenergy production are measured as the carbon emissions from the burning of woody biomass net the reduction in carbon emissions due to fossil fuel substitution. The changes can be represented by the parameter η in our numerical model (see Equation 13). η is originally defined as the proportion of carbon that is permanently stored in finished wood

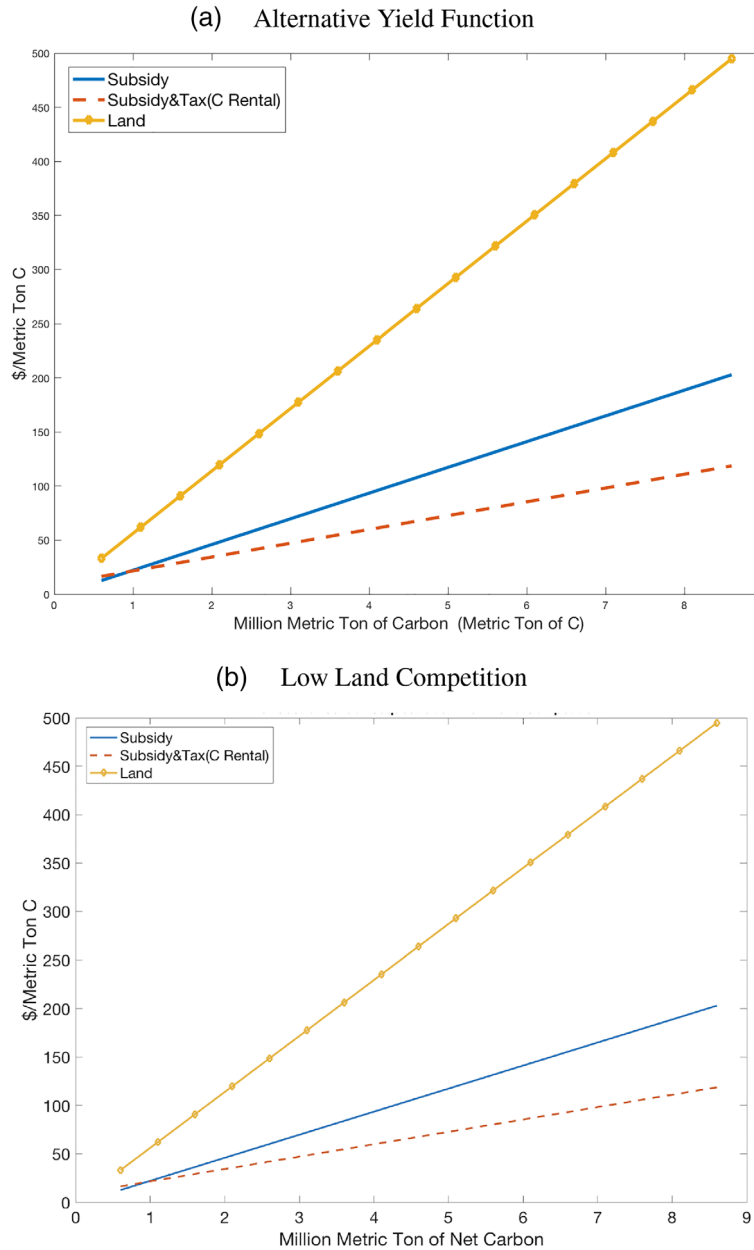

FIGURE 12 Marginal cost curves-additional robustness checks

Notes: Figure 12 Depicts the marginal cost curves under two alternative assumptions. (a) Assumes the yield function used in Cunha-e-Sá et al. (2013), which is displayed in Figure 11; (b) assumes that the price elasticity of land supply $\varepsilon = 1$; the results of subsidy and tax and C rental are identical so we combine their results

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products after the stand is harvested (i.e., the pickling rate). In the case of woody bioenergy use, η can also represent the extent of reduction in carbon emissions from fossil fuel consumption. For robustness checks, we test alternative assumptions of η. In the reference scenario, we assume that η equals 0.35 in the reference scenario (Daigneault et al., 2010). If the substitution for fossil fuel is 100%, then $\eta = 1$ (for example, carbon capture and storage with bioenergy production); if there is no substitution, then $\eta = 0$. For change (2) above, we test a high demand function, which is displayed in Figure 9. All the robustness checks of our results to these additional assumptions (see Figure 10) suggest that the inclusion of woody bioenergy production does not change our findings on the relative marginal costs of mitigation for different carbon policies.

6.3 | Alternative yield functions

The choice of yield function may affect the result, because the yield function can affect the rate of carbon accumulation at the intensive margin. The analysis in this study is based on US southern pine, which is a typical tree species in temperate regions. We also test a tropical species used in the study of Cunha-e-Sá et al. (2013), which has a much younger age range (Figure 11). The results in our study are generally consistent with the previous findings except that the differences between per ton and per hectare are smaller with respect to the earlier yield function, because the allowable age range is now limited (see Figure 12a). Cunha-e-Sá et al. (2013) show that, compared with a per-ton policy, a per-hectare policy results in very marginal welfare losses (0.42% to 5.56%). However, our results reveal that the magnitude of differences in the marginal costs of carbon sequestration between per-hectare and per-ton policies is considerably larger under the same yield function. Given the same carbon sequestration targets, the land subsidy policy is two to four times more expensive than the subsidy and tax and C rental policies.

6.4 | Low land competition

Figure 12b shows the results of the case of higher elasticity of land supply ($\varepsilon = 1$) to reflect low land competition. The results are consistent with previous findings: The marginal cost curve of the land subsidy is higher than that of the tax and subsidy scenario, and diverges at a high carbon price. However, when land supply becomes more elastic, there is a smaller difference between subsidy and tax and land subsidy. The marginal costs of carbon sequestration of land subsidy is two to four times more expensive than those under subsidy and tax. In general, the results indicate that per-hectare policies (land subsidy) still result in significantly higher efficiency losses than per-ton policies (subsidy and tax, C rental, and carbon subsidy) at each level of forest sequestration.

7 | CONCLUSION

In this study, we measure the efficiencies of different policies to promote carbon sequestration by considering the policy-induced market effects and intertemporal changes at both the intensive margin and the extensive margin. We investigate the dynamics of the effects of different policies on forest inventory, the timber market, and carbon consequences. The intertemporal optimization model is employed to compare the marginal costs of sequestration for alternative policies under different land-use conditions, yield functions, and carbon price paths. We find that a carbon tax on forest emissions results in net carbon emissions, because it ignores the benefit of the carbon stored in trees as they grow. The tax-only policy leads to losses of forestland, as the tax on carbon reduces expected land value in forests compared to other uses. Moreover, the tax-only policy fails to promote carbon

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sequestration at the intensive margin, because the positive effects on harvests are small or even negligible.

The per-hectare policy cannot provide adequate incentives to sequester more carbon at the intensive margin. Our study shows that the efficiency differences between the per-hectare policy and the per-ton policy could be substantially larger than recent numerical studies. The per-hectare policy may be 2–16 times more expensive than the two first-best policies (carbon subsidy and tax and C rental). Our cost estimates of the per-hectare policy are also substantially higher than the ranges of existing estimates in the carbon sequestration cost of forests, which do not account for policy choices. In addition, the subsidy-only policy is much more efficient than the per-hectare policy. Among all non-optimal policies, the subsidy-only policy incurs costs similar to those incurred by the two first optimal policies and, therefore, is the second optimal option. Moreover, a higher carbon price increases the differences in costs between the per-hectare policy and the optimal policies.

This study provides practical implications for carbon policy design in the land-use sector. First, when designing climate policies, it is recommended that the land-use sector is treated differently from other emission sectors. Although it is correct to tax carbon emissions from energy use directly, taxing emissions from the forest sector without compensating for sequestration leads to inefficient outcomes, because the carbon tax reduces the expected land value and the rate of carbon sequestration in forests. Second, policy choices should be considered as an important factor contributing to differences in the costs of carbon sequestration. Forest carbon sequestration may be too expensive if the selected policy cannot induce more carbon gain at the intensive margin. It is worthwhile to implement a per-ton policy, rather than a per-hectare policy, provided that the administration costs do not exceed the efficiency losses. Second, a per-ton carbon subsidy policy may be an appropriate choice if the measurement costs are considered, because this policy provides efficiency that is closely approximated to that of the optimal policies and avoids the measurement costs of actual emissions due to harvests.[16]

Our findings also shed light on the potential carbon policies regulating woody biomass production. Woody bioenergy is conventionally treated as carbon neutral (US Environmental Protection Agency, 2011). Following this viewpoint, the policy suggestion is to subsidize carbon sequestration from growth but not to tax carbon emissions from burning wood bioenergy, which is the carbon subsidy scenario in our analysis. However, there is a significant debate over whether wood used for energy is carbon neutral. Some authors suggest that emissions from wood-based biofuels should be treated differently than emissions from non-wood-based biofuels (e.g., Searchinger et al., 2009). Similarly, life cycle analyses have concluded that carbon emissions from wood-based biofuel products should be taxed (e.g., Manomet Center For Conservation Sciences, 2010). Both these studies suggest that the burning of long-established forests causes large releases of carbon, and as a result, the authors suggest that the policy remedy is to tax carbon emissions generated by wood-based biofuels (tax-only scenario). According to the findings in our study, their policy recommendations could lead to inefficient outcomes and net carbon emissions, because the carbon tax reduces the rate of carbon sequestration in forests.

The generalizability of our results is subject to certain limitations. For instance, one assumption with the forest carbon policies is that the non-forestland use (e.g., agricultural land) is not taxed or regulated by carbon policies, which future research could relax. Another limitation is that we consider only the harvest age at the intensive margin, but more management practices (e.g., thinning and fertilizing) need to be taken into account in future studies. Moreover, more future work needs to be done to examine the indirect regulations of carbon sequestration beyond the direct regulations (Favero & Mendelsohn, 2014).

[16]The carbon emissions related with the forest sector are either from harvests and processing of wood products or from the forest sites, such as dead trees, branches, leaves, understory, and forest floor. The first part is determined by the proportions of different types of wood products in the market and how they are consumed and recycled in their life cycles. The latter part is a natural process but can also be disturbed by human or climate factors. Both parts could be very uncertain and difficult to monitor. Nonetheless, the carbon sequestration is a result of the process of tree growths, which is measurable and predictable.

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